TECHNICAL DESIGN DOCUMENT (TDD)

BrokerLens: Customs Broker AI Document Extraction & 3-Way Cross-Check Workbench | Problem Statement #3

Project	BrokerLens — Automated Customs Document Reconciliation Workbench
GitHub Repository URL	**https://github.com/Ojaswi-Gupta/broker_lens.git**
Live Deployment / URL	http://localhost:3000 (Local / Docker Compose) | Cloud Ready (AWS ap-south-1 Mumbai)
Target Client	Company A, Navi Mumbai (Freight Forwarder & CHA operating at JNPT / Nhava Sheva Port, INNSA)
Author & Version	Ojaswi Gupta | Production Release v1.0.0 (November 2026)
Project Sizing Bounds	Budget: ■**7.5 Lakh Max** | Delivery: **6 Weeks Max** | Target SLA: **< 15 min entry &** ≥ **95% accuracy**

1. Executive Summary & Quantified Client ROI Model

Company A is a prominent customs broker handling **1,100 import consignments per month** (13,200/year) at **JNPT Nhava Sheva (INNSA)**. A 14-member documentation executive team manually verified Commercial Invoices, Packing Lists, and Bills of Lading, causing severe operational bleed.

Cost Vector / Metric	Baseline (Without BrokerLens)	With BrokerLens	Net Annual Savings
1. Manual Data Entry Cost	**■34.65 Lakh / yr** (9,900 hrs @ ■350/hr; 45m/ship)	**■8.78 Lakh / yr** (2,508 hrs @ 11.4m/shipment)	**■25.87 Lakh / yr (7,392 staff hrs saved; -74.7%)**
2. Demurrage & BoE Amendments	**■35.64 Lakh / yr** (396 cases @ 3.0% × ■9,000)	**■9.11 Lakh / yr** (101 cases @ 0.77%; 74.5% cut)	**■26.53 Lakh / yr (295 error cases prevented)**
TOTAL ANNUAL PROBLEM COST	**■70.29 Lakh / yr**	**■17.89 Lakh / yr**	**■52.40 Lakh / yr Net Savings (698% First-Year ROI on ■7.50L budget)**

2. Explicitly Labeled Assumptions (ASSUMPTION-01 to ASSUMPTION-06)

`ASSUMPTION-01`	**Customs Flat-File Layout**	Company A keeps their customs filing desktop software (VisualIMPEX/ICEGATE). BrokerLens outputs a 27-column UTF-8-BOM CSV and TSV clipboard format matching Nhava Sheva import conventions.
`ASSUMPTION-02`	**Calibrated Confidence Tiers**	Confidence ≥ 85% is High (Emerald); 45%–84% is Low Confidence (Amber) requiring explicit confirmation; < 45% is Unreadable (Rose) routing to Guided Manual Entry.
`ASSUMPTION-03`	**Zero-Tolerance Cross-Checks**	Invoice line math allows ≤ 0.01 rounding delta. Gross Weight, Net Weight, Packages, Qty, and Container Number across Invoice, PL, and BoL require exact match (0.00 kg tolerance by default).
`ASSUMPTION-04`	**RBAC & Override Escalation**	Executives can edit fields and sync cross-checks. Overriding a failed cross-check with an audit note escalates the consignment to require Documentation Manager sign-off (HTTP 403 for Execs).
`ASSUMPTION-05`	**Hybrid Vision AI & Fallback**	Google Gemini 2.5 Flash Vision with Zero Data Retention headers, backed by an embedded deterministic pypdf + layout parser for 100% offline testability without paid API keys.
`ASSUMPTION-06`	**DPDP Act 2023 & Data Residency**	Hosted in AWS Mumbai (ap-south-1). Importer contact name, phone, email, and IEC support live redaction via the DPDP PII masking engine.

3. System Architecture & Relational Persistence Schema

BrokerLens decouples the presentation tier from business logic and database persistence: **Next.js 14 App Router** (TypeScript, Tailwind) + **FastAPI** (Python 3.10+, SQLAlchemy 2.0, Pydantic v2) + **PostgreSQL 16 / SQLite**.

Table Name	Primary Key & Foreign Keys	Key Columns & Role in Pipeline
`users`	id (PK)	email, hashed_password, role (DOCUMENTATION_EXECUTIVE, DOCUMENTATION_MANAGER), full_name
`shipments`	id (PK), approved_by_id (FK)	shipment_code (SHP-24811), supplier_name, importer_iec, status, requires_manager_approval, exported_at
`shipment_documents`	id (PK), shipment_id (FK)	doc_type (INVOICE, PACKING_LIST, BILL_OF_LADING), file_url, skew_angle_deg, is_unreadable, avg_confidence
`extracted_fields`	id (PK), shipment_id (FK)	doc_type, field_key, ai_extracted_value, current_value, confidence_score, is_confirmed_by_human, bbox_hint
`shipment_line_items`	id (PK), shipment_id (FK)	line_number, description, hsn_code, qty, unit_price, line_total, gross_weight_kg, is_confirmed_by_human
`crosscheck_results`	id (PK), shipment_id (FK)	rule_code, rule_name, invoice_value, packing_list_value, bol_value, discrepancy_delta, status (PASSED/FAILED)
`field_edit_history`	id (PK), shipment_id (FK)	shipment_code, field_key, original_ai_value, updated_human_value, was_ai_correct, edited_by_role, edit_reason

4. The 6-Rule Deterministic 3-Way Cross-Check Engine

`INVOICE_TOTAL_VS_LINE_SUM`	line_total = round(qty × unit_price, 2); sum(lines) == invoice_total (delta ≤ 0.01)	**CRITICAL**	Blocks export on line arithmetic or sum mismatch
`THREE_WAY_GROSS_WEIGHT`	Gross_Wt(Invoice) == Gross_Wt(PL) == Gross_Wt(BoL) [exact match]	**CRITICAL**	Intercepts SHP-24811 45 kg delta; blocks export
`TWO_WAY_NET_WEIGHT`	Net_Wt(Invoice) == Net_Wt(PL) and Net_Wt < Gross_Wt (physical sanity check)	**WARNING**	Flags tare weight omission or negative dunnage
`THREE_WAY_PACKAGES`	Packages(Invoice) == Packages(PL) == Packages(BoL)	**CRITICAL**	Detects carton count mismatch across carriers
`TWO_WAY_ITEM_QTY`	sum(Invoice_Lines.qty) == Total_PL_Qty	**WARNING**	Verifies piece count consistency per tariff line
`THREE_WAY_CONTAINER`	Normalized(Container_No_Inv) == PL == BoL and Shipper match	**CRITICAL**	Blocks misrouted containers at port gates

5. Vision AI Pipeline, Confidence Calibration & Edge Cases

Seed Consignment	Supplier & Document Anomaly	Automated Detection & Handling Flow
`SHP-24811`	Company B (China): Invoice states 3,120 kg; PL & BoL state 3,165 kg (bilingual ■■■).	Cross-check engine catches 45 kg delta; blocks export (HTTP 422). 1-click sync or manager-gated override unlocks filing.
`SHP-24812`	Company C (Sweden): -2.8° scanner tilt; Swedish FAKTURA compound headers.	Viewer toolbar provides 1-click Auto-Deskew (+2.8°); flags Incoterm (78%) and HSN (74%) in Amber for human confirmation.
`SHP-24813`	Shenzhen Precision Motors: Clean digital invoice, PL, and BoL.	All 6 cross-checks pass (100%); zero blockers; unlocks immediate 1-click ICEGATE 27-column CSV export.
`SHP-24814`	Hamburg Chemie GmbH: Degraded thermal scan noise (avg confidence 31%).	Flags document as UNREADABLE; initiates Guided Manual Entry fallback mode with side-by-side visual reference.

6. UI & Design System Rationale (The 7 Screens)

Screen 1: Login /login	Role portal with 1-click demo role switchers (Documentation Executive Rohan vs Manager Priya) and DPDP compliance notices.
Screen 2: Queue /shipments	Real-time operational dashboard with 4 KPI cards, tabbed status filters (Failed, Review, Approved), and full-text search.
Screen 3: Upload /shipments/new	3-document drag-and-drop ingestion with 15MB file validation + 1-Click Scenario Runner for evaluators.
Screen 4: Viewer /shipments/[id]/review	50/50 split canvas. Left side features interactive PDF canvas with zoom, pan, and manual de-skew controls (-5° to +5°).
Screen 5: Review /shipments/[id]/review	Side-by-side 3-way cross-check matrix with visual discrepancy deltas, 1-click sync buttons, and audit trail drawer.
Screen 6: Export /shipments/[id]/export	Pre-flight gatekeeper with live HTTP 422 test button. Cleared state provides 27-col ICEGATE CSV with UTF-8 BOM & TSV copy.
Screen 7: Analytics /analytics	Manager SLA report (95.8% accuracy vs 95% target), Problem Statement #3 ROI formula breakdown, and immutable edit history.

7. Scope Trade-Offs & Post-MVP Roadmap

What We Cut: (1) Direct ICEGATE EDIFACT message dispatch (requires customs DSC token and dedicated VPN tunnel; replaced by 1-click 27-col flat-file CSV). (2) Automated HSN classification/duty calculation (avoids penalty risk under Customs Act Sec 111(m); kept human-in-the-loop). (3) External importer portal (focused 100% of 6-week engineering effort on CHA operations).

Next 6-Week Roadmap: Sprint 7–8: IMAP/Email automated attachment ingestion into the upload queue. Sprint 9–10: Supplier-specific few-shot prompt injection driven by historical edits in FieldEditHistory. Sprint 11–12: Multi-page cross-table pagination stitching for 100+ line item industrial machinery consignments.